

Bradley Lignoski

Computer Science Faculty | AI & Machine Learning | Cloud Computing | Full-Stack Engineer

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PROFESSIONAL SUMMARY

Computer Science educator and software engineer with over a decade of classroom teaching—including 12 years teaching CS and mathematics in Bellevue, WA—plus full-stack engineering at a high-growth Seattle tech company. Master of Science in Artificial Intelligence & Machine Learning (4.0 GPA, Johns Hopkins, May 2026) with graduate concentration in machine learning, deep learning, cloud-deployed AI systems, and data science. Prepared to teach a variety of undergraduate CS courses spanning programming, algorithms, data structures, software engineering, machine learning, artificial intelligence, deep learning, cloud computing, web programming, database systems, and data science (see *Course Teaching Readiness* below). Committed to inclusive, equity-centered pedagogy and to connecting foundational CS to industry-grade practices.

EDUCATION

Master of Science, Artificial Intelligence & Machine Learning

Feb. 2024 – May 2026

Johns Hopkins University, Whiting School of Engineering

4.0 GPA

- **Coursework:** Algorithms for Data Science (A+), Applied Machine Learning (A), Artificial Intelligence (A+), Creating AI-Enabled Systems (A), AI/ML Ops (A+), Data Visualization (A+), Logic: Systems, Semantics & Models (A), Production AI Systems (Cloud) (A), ChatGPT from Scratch (A+), Mechanistic Interpretability (A).
- **Algorithms & Data Structures:** runtime complexity analysis, complexity classes, graph algorithms, state-space search, local search, adversarial search, constraint satisfaction, optimization, dynamic programming, and algorithmic proofs.
- **Machine Learning, Deep Learning & AI:** supervised/unsupervised learning, model evaluation (precision, recall, F1, ROC), regularization, hyperparameter tuning; neural architectures (MLPs, CNNs, RNNs, transformers); built a GPT-style language model end-to-end; mechanistic interpretability of transformer internals (linear probes, sparse autoencoders, activation/path patching).
- **Cloud Computing & AI Systems:** production AI on Microsoft Azure via Terraform (IaC), Docker/Flask, CI/CD; full-lifecycle AI system design; built AI systems for fraud detection, facial recognition, object detection (YOLO), and recommendation engines; knowledge graphs (OWL2, SPARQL).

Bachelor of Arts — Physics (Minor: Mathematics)

Fall 1999 – Spring 2003

The College of Wooster — Wooster, OH

- Rigorous technical degree with coursework in programming, linear algebra, differential equations, probability & statistics, vector and tensor calculus, and computational methods; independent study thesis on quantum decoherence.
- **Undergraduate Teaching Assistant**, Introductory Physics Labs (PHYS 20300/20400), Spring 2001–Spring 2003: hands-on equipment setup and troubleshooting, data analysis, theory.

UNDERGRADUATE COURSE TEACHING READINESS

- **Intro CS, Data Structures & Algorithms:** 5 yrs as sole CS instructor at Hillside teaching mixed-level Python programming plus electives including Analysis of Algorithms; built CS curriculum from scratch at two schools; JHU Algorithms for Data Science (A+), Artificial Intelligence (A+).
- **Programming Languages & Software Engineering:** Python (15+ yrs), JavaScript/TypeScript, SQL, $\text{\LaTeX}$; 1.5 yrs full-stack at Convoy (React, Node.js, Apollo, microservices, CI/CD, Pytest, agile).
- **Machine Learning:** JHU Applied ML (A), AI/ML Ops (A+), Creating AI-Enabled Systems (A); PyTorch, Scikit-Learn; model evaluation, drift detection, hyperparameter tuning, MLOps.
- **Artificial Intelligence:** JHU Artificial Intelligence (A+): search, optimization, constraint satisfaction, reasoning, planning, reinforcement learning; AI ethics (fairness, bias, responsible deployment).
- **Deep Learning:** JHU ChatGPT from Scratch (A+) — built a GPT-style transformer end-to-end; JHU Mechanistic Interpretability (A) — transformer internals, attention/MLP circuits, sparse autoencoders.
- **Cloud Computing:** JHU Production AI Systems (A) — deployed AI services on Microsoft Azure via Terraform Infrastructure-as-Code, containerized with Docker.

- **Web Programming & Database Systems:** Full-stack React/Apollo/Node.js at Convoy; React Native + Firebase + Google Cloud Functions; Khan Academy interactive problem sets; SQL backfill of 11M production records.
- **Data Science:** JHU Algorithms for Data Science (A+), Data Visualization (A+); Pandas, NumPy, Matplotlib, Seaborn; statistical modeling and exploratory analysis.

TEACHING & CURRICULUM EXPERIENCE

Math Department Chair & CS/Math Teacher

Dec. 2005 – June 2017

Hillside Student Community — Bellevue, WA

- Built the school's Computer Science program from scratch as sole instructor; taught mixed-level Introductory CS and CS electives including Analysis of Algorithms alongside Honors Precalculus (with optional Python projects), Calculus AB/BC, and Lab Physics over a 12-year tenure—routinely delivering college-level material to advanced 10th–12th graders.
- Developed full CS curricula in Python (Replit, Pytest) with project- and inquiry-based programming, automated testing, and formative/summative assessment; recorded hundreds of CS and Precalculus video lessons.
- Held daily one-on-one and small-group sessions—including during lunch periods—to diagnose root-cause conceptual gaps and rebuild foundations before reintroducing course material; served students with diverse learning differences using differentiated instruction and universal-design strategies.
- As Math Department Chair, led a backward-design audit of the math curriculum—working back from Algebra II and Precalculus to identify lower-grade gaps and low-value content to cut (drawing on my Common Core work at Khan Academy); mentored colleagues and led textbook and assessment selection.
- Co-led math faculty hiring—designed a technical-interview-style mock-tutoring evaluation; co-observed and deliberated on candidate demo lessons with the school director.

Mathematics & Computer Science Instructor

Sept. 2019 – June 2021

The Northwest School — Seattle, WA

- Designed and taught the school's first formal introductory CS course (online/hybrid through the pandemic) on Replit with automated tests for instant student feedback, plus live screen-share sessions.
- Co-led a three-teacher revision of the Precalculus curriculum, authoring quizzes, tests, recorded video lectures with embedded learning-check mini-quizzes, homework, and full solution guides for student self-review; taught three Precalculus sections and a transitional Mathematical Analysis course bridging Algebra II and Precalculus.
- Held weekly one-on-one office hours and scheduled additional sessions timed for international students in China—an equity-centered, culturally responsive practice ensuring no student lost access due to time zone.

Private Math Tutor

2017 – 2020

Independent — Bellevue, WA

- One-on-one in-person tutoring of high-school students, Algebra II through AP Calculus BC; individualized lesson plans diagnosing root-cause conceptual gaps before rebuilding to course material.

Curriculum Developer

Sept. 2013 – Mar. 2014

Khan Academy — Mountain View, CA (Remote)

- Built hundreds of interactive online problem sets (JavaScript + $\text{\LaTeX}$), now used by ~ 3 million users (e.g., Math 3 course); authored Python scripts that eliminated 100% of targeted errors in the peer-review pipeline.

SOFTWARE ENGINEERING EXPERIENCE

Full-Stack Software Engineer

Sept. 2021 – Feb. 2023

Convoy Inc — Seattle, WA

- Built contract-negotiation tooling for 400,000+ clients across three React apps and multiple Apollo/Node.js microservices (Agile); closed unit-test gaps and resolved 200+ bugs.
- Implemented an advanced AG-Grid hotkey system that reduced account-manager task time by 99%; authored a batched SQL backfill of 11 million mission-critical records, unblocking a data-science initiative.

RESEARCH

Transformer Mechanistic Interpretability (JHU, 2025–2026): per-feature causal pipeline (activation + path patching on Qwen3-0.6B-Base) testing cross-layer feature persistence—copy-forward vs. distributed recomputation.